

ROHAN THAKUR

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PROFILE

AI/ML engineer specializing in RAG pipelines, agentic LLM systems (LangGraph/LangChain), and end-to-end ML deployment. Built production retrieval systems with pgvector, multilingual GPT-4o backends, and a self-critiquing autonomous agent.

EDUCATION

Graphic Era Hill University, Dehradun	Aug 2022 – June 2026
B.Tech CSE (Hons.) in AI & ML	CGPA: 7.33
Mount Litera Zee School, Roorkee	2022
Senior Secondary (XII)	73.8%

EXPERIENCE

AI/ML Intern	Jan 2026 – Present
AI Future Gen Pvt. Ltd., Dehradun	
- Built a RAG pipeline in Node.js over 30+ PDF/JSON documents (1,000-char chunks, 200-char overlap), indexed in PostgreSQL/pgvector via HNSW cosine search on 1,536-dim embeddings. - Added session-aware conversational memory with PostgreSQL chat history, feeding recent messages plus top-8 retrieved chunks into GPT-4o-mini with per-query similarity scoring. - Engineered a multilingual backend (Node.js + Whisper & GPT-4o-mini) converting voice/text in any language into structured JSON rules for SMS filtering.	

PROJECTS

Autonomous AI Newsletter Agent (Live)	LangGraph, LangChain, Groq, Tavily, FastAPI, Docker
Live Demo: Newsletter Agent App GitHub: Repository	
- Built a 7-node LangGraph state-machine agent (plan → research → summarize → draft → critique → approval → send) with a self-critique revision loop and LLM-scored conditional routing. - Implemented Human-in-the-Loop pause/resume via LangGraph's <code>interrupt_before</code> and <code>MemorySaver</code>, keyed by per-session <code>thread_id</code> so approval state survives across independent HTTP requests. - Designed a FastAPI backend with Server-Sent Events streaming live per-node trace; integrated Groq Llama-3.3-70B and Tavily for ~45s end-to-end runs. - Containerised with Docker and deployed on Hugging Face Spaces with secrets injected via Space environment variables.	
AI-Powered Fraud Detection & Prevention Ecosystem with RPA	XGBoost, SHAP, FastAPI, React, SMOTE, RPA
GitHub: Repository	
- Benchmarked XGBoost, LightGBM, Random Forest, and AdaBoost on the IEEE-CIS Fraud dataset (590K+ transactions, 471 features); XGBoost achieved F1 = 0.6762, Precision = 0.89, AUC-ROC = 0.9522. - Addressed 3.5% class imbalance with SMOTE applied post-split to prevent data leakage; integrated SHAP TreeExplainer for per-transaction top-5 feature attributions, surfaced in a React 19 analyst dashboard with FastAPI backend (<2s latency). - Engineered a 4-bot Python RPA pipeline (customer alert, risk-routed case creation, RBI-format STR generation, audit logger) auto-triggered on fraud probability >0.5, cutting per-case manual workflow from ~12 min to <2 s.	
AI Traffic Sign Detection	TensorFlow, Keras, CNN, Flask
Live Demo: Traffic Sign Detection App GitHub: Repository	
- Developed a CNN-based deep learning model using TensorFlow and Keras achieving 98%+ accuracy across 43 traffic sign classes. - Built a Flask-based inference API and web interface enabling real-time traffic sign classification. - Deployed the model behind a Flask API with a simple web UI on Hugging Face Spaces; anyone can upload a sign image and get a prediction in real time.	
Personal Finance Tracker (Live)	React.js, Chart.js, Tailwind CSS
Live Demo: personal-finance-tracker.app GitHub: Repository	
- Developed and deployed a finance tracking app with interactive Chart.js dashboards for visualizing spending patterns and budget. - Used React Context API for state management and deployed on Vercel — currently live and functional.	
TECHNICAL SKILLS	
AI/ML & GenAI: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, LLMs, RAG, LangGraph, LangChain, Embeddings, Vector Databases (pgvector), Pandas, NumPy	
Languages: Python, JavaScript, C++, SQL	
Backend & Infrastructure: FastAPI, Flask, Node.js, REST APIs, Server-Sent Events, PostgreSQL (pgvector), Docker	
Frontend: React.js, Tailwind CSS	
Deployment & Tools: Hugging Face Spaces, Docker, Git/GitHub, Power BI, Kaggle, RPA (Python bots)	
Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, OOP, DBMS, SDLC	

CERTIFICATIONS

Professional Machine Learning Engineer — Google Cloud	Jul 2024
Microsoft Certified: Power BI Data Analyst Associate — Microsoft	May 2025
Machine Learning with Python — SLOG Solutions Pvt. Ltd.	Nov 2023